

Dual-Track Traversal & Ambiguity Tiebreak

Track 3a: Lossless Prompt Compression

- Encodes complete evidence multigraph G_q^* losslessly
- Shared passage dictionary $\mathcal{S}_p$ eliminates text redundancy
- **100% evidence recall** (retains all row attributes)

Track 3b: Grounded Path Enumeration

Bounded DFS enumerates non-filtering reading guides:

Path p_1 : Toronto Eagles $\xrightarrow{\text{rec}}$ ρ_0 $\xrightarrow{\text{city}}$ Toronto $\xrightarrow{\text{link}}$ Doc $\xrightarrow{\text{text}}$ York

- Multi-factor Score: $S(p_1) = 7.85$ (Rank 1)

Path p_2 : Hamilton Tigers $\xrightarrow{\text{rec}}$ ρ_1 $\xrightarrow{\text{city}}$ Hamilton ...

- Multi-factor Score: $S(p_2) = 3.10$ (Year mismatch)

if score collision

Track 3c: Structural Ambiguity Tiebreak

When candidate entities tie on relational graph scores:

- Collision condition: $|\mathcal{C}_q| > 1 \implies \text{Flag SCORE_COLLISION_TIE}$
- Bridge lexical tiebreak score over connected passages:

$$\tau(c) = \sum_{w \in \text{Tokens}(c) \cap \mathcal{D}^{\text{bridge}}} \text{IDF}(w)$$

- Prioritizes candidate ρ_0 to prevent arbitrary hallucination

Evidence Multigraph

Suggested Paths

Tiebreak Winner

Unified Prompt Payload & LLM Generation

Serialized Unified Prompt Payload:

[SYSTEM]: Answer with exact span from verified evidence.

[GRAPH EVIDENCE]:

Record_0: {Club: "Toronto Eagles", Formed: "1989", City: "Toronto"}

Record_1: {Club: "Hamilton Tigers", Formed: "1991", City: "Hamilton"}

[SHARED PASSAGES]:

Doc:Toronto: "Originally established in 1793 as the town of York..."

[SUGGESTED READING GUIDE]:

Path 1 (Score 7.85): Toronto Eagles $\rightarrow$ Record_0 $\rightarrow$ Toronto $\rightarrow$ York

[AMBIGUITY RESOLUTION]: Prioritized Candidate: Record_0

[QUESTION]: Original name of city with AFL club formed in 1989?

Unified Context

Backbone LLM Forward Pass ($T = 0$)

Single-turn in-context generation conditioned on complete multigraph context and guided by non-filtering candidate paths.

Greedy Decode ($T = 0$)

Grounded Answer: "York"

Auditable Provenance Record:

- Table: AFL_Ontario_0
- Row: 0
- Cell: City: Toronto
- Passage Bridge: /wiki/Toronto
- Target Span: "York"